Newton's comparison: areas, action defects and controlled reunion

For horizontal speed $v > 0$ and downward force $F > 0$, Galileo's inertial line and falling parabola enclose area $A = vF\tau^3/(6m)$, with the exact conversion

$$\boxed{(3F/v)A = \tau\Delta E, \qquad \Delta E = F^2\tau^2/(2m).}$$

The matched-endpoint chord lens is half that area and has action defect $F^2\tau^3/(24m) = \tau\Delta E/12$. A genuine receding-centre limit preserves this chord defect but cancels the apparent tangent-sector remainder. A specified $F, -F, F$ force loop reunites both position and velocity and gives an exact scalar quantum phase for every common motional preparation, once quantum kinematics and coherent control are supplied. None of these geometric or phase statements establishes independent positive-action necessity.

This treatment integrates the constant-force derivation and its receding-centre correction with the controlled reunion. Written model/coordinator checks are internal; the mechanics and phase ingredients are established and no literature-priority claim is made.

The source geometry and three different areas

Take $m, F, v, \tau > 0$, downward y, and

$$q_I(t) = (vt, 0), \qquad q_F(t) = (vt, Ft^2/(2m)), \quad 0 \le t \le \tau.$$

The inertial–fall region is closed by the vertical segment at $x = v\tau$. Direct integration gives $A = vF\tau^3/(6m)$, sagitta $s = F\tau^2/(2m)$ and deterministic transverse kinetic gain $\Delta E = Fs = F^2\tau^2/(2m)$. Total mechanical energy is conserved; the potential loses that amount. This ΔE is not a quantum energy standard deviation.

Region	Positive ordinary area
Inertial horizontal line to falling parabola	$vF\tau^3/(6m)$
Matched-endpoint chord to falling parabola	$vF\tau^3/(12m)$
Inertial line to endpoint chord	$vF\tau^3/(4m)$

The first is twice the second; the third is their sum. Kepler sectors about a force centre are another construction. A horizontal Galilean boost to speed v removes the plotted spatial area, while a matched-endpoint action difference remains: the common horizontal action cancels. The conversion therefore specifies a frame with $v \neq 0$.

Diagram of constant-force regions

The read witnesses are Motte/Chittenden 1846, Definitions I–VIII and Scholium, Laws/corollaries, Book I Section I and Section II Propositions I–II, and the Motte 1729/Wilkins excerpt. Reading level: the named English passages, not every edition or full historical literature. Their relevant paragraph map is:

Passage, 1846 printed pages	Use
Definitions II, IV, VII–VIII, 73–77	Momentum and impressed, accelerative and motive force
Definition Scholium, 77–82	Mathematical time/space versus sensible measures
Laws Corollaries V–VI, 89	Relative motion under common translations/accelerations

Passage, 1846 printed pages	Use
Laws Scholium, 89–90	Projectile parallelogram and Galileo's parabola
Lemmas I–III, 95–96	Limits of inscribed/circumscribed figures
Lemma X, 99–100	Initial force displacement quadratic in duration
Lemma XI Cor. 4–5, 101	Cubic tangent triangles and one-third curved segment
Section I closing Scholium, 102–103	Evanescent divisible quantities and limiting ratios
Proposition I, 103–105; II, 105–106	Central impulse polygon and converse area-direction law

The common parallel force belongs to the projectile Scholium and Lemmas X–XI; Proposition I assumes a fixed centre. Some source equations are images; Wilkins's PDF and the projectile and central polygon figures supplied the recorded passage checks. The historical limits are not evidence for an established fluxional-time interpretation or a physical $\hbar \to 0$ theorem.

Matched endpoints, arbitrary partitions and action normalization

For $L = m|\dot{q}|^2/2 + Fy$, the same-endpoint chord has $y_C(t) = F\tau t/(2m)$. If $y = y_F + \eta$ and both endpoint variations vanish, integration by parts using $m\ddot{y}_F = F$ gives

$$S[y_F + \eta] - S[y_F] = \frac{m}{2}\int_0^\tau \dot{\eta}^2 dt.$$

For $\eta = Ft(\tau - t)/(2m)$ this is

$$D_{C,F} = \frac{F^2\tau^3}{24m} = \frac{F}{2v}A_{\text{chord,fall}} = \frac{F}{4v}A_{\text{inertial,fall}} = \frac{\tau\Delta E}{12}.$$

A common ordinary point-function term $dG(q,t)/dt$ added to L cancels because the endpoints and times agree. A velocity-dependent G changes the endpoint data required by that statement: adding $d(\kappa\dot{y})/dt$ changes this chord-minus-parabola action by $-\kappa F\tau/m$, since their endpoint velocities differ. That is a higher-derivative Lagrangian, outside this point-mechanics convention. The bare inertial and falling paths have different vertical endpoints; an observable phase comparison needs endpoint/control phases.

For a partition $0 = t_0 < \cdots < t_N = T$ with steps τ_j, the polygon interpolating exact trajectory values has

$$D_\pi = \frac{F^2}{24m}\sum_j \tau_j^3 \le \frac{F^2 T}{24m}|\pi|^2 \longrightarrow 0, \qquad A_\pi = \frac{vF}{12m}\sum_j \tau_j^3.$$

The local tangent slope cancels in each chord lens. At fixed N convexity minimizes the sum at equal steps, $\sum \tau_j^3 = T^3/N^2$. Splitting a cell into $a, b > 0$ decreases its defect by $F^2ab(a+b)/(8m)$. These are sampled chords; an executed force-impulse polygon has its own dynamical error estimate. The unmarked vertex impulse $F\tau_j$, sagitta and area can all vanish while $2ms(\tau_j)/\tau_j^2 = F$ stays exact.

Neighbouring off-shell variations $\eta = \alpha t(\tau - t)$ have defect $m\alpha^2\tau^3/6 \to 0$. Positive Dirichlet curvature does not make a smallest positive defect. Likewise the deterministic work ΔE is not the σ_H in Mandelstam–Tamm:

$$\sigma_H\sigma_B \ge \tfrac{1}{2}|\langle[H,B]\rangle|, \qquad \tau_B = \frac{\sigma_B}{|d\langle B\rangle/dt|}, \qquad \sigma_H\tau_B \ge \hbar/2,$$

under the proper domains and a nonzero denominator. With $v < c$, the Newtonian constant-force segment stays below c if $\tau < (m/F)\sqrt{c^2 - v^2}$; arbitrarily small intervals remain possible. The foundations manuscript retains its distinct relativistic variation calculation. Feynman's least-action discussion, passage, uses contributions with action differences within $\hbar$, not a prohibition on such paths. A supplied phase rule $e^{iS/\hbar}$ gives period $h = 2\pi\hbar$ and an order-one comparison duration $(24m\hbar/F^2)^{1/3}$ for this pair; it does not select $\hbar$.

A genuine receding centre: subtract before taking the limit

For $C_R = (0, R)$ define signed sector area

$$A_R[\gamma] = \frac{1}{2}\int_\gamma [x\,dy - (y - R)\,dx] = \frac{R}{2}\Delta x + \frac{1}{2}\int_\gamma (x\,dy - y\,dx).$$

The R term cancels exactly for paths sharing endpoints, giving their closed loop area. For a fixed limiting parabola its finite part is $Fv\tau^3/(12m)$. A varying family must also retain its endpoint correction. Use the actual central-force solutions

$$m\ddot{q}_R = (F/R)(C_R - q_R), \quad q_R(0) = 0, \quad \dot{q}_R(0) = (v, 0), \quad \omega_R^2 = F/(mR),$$
$$x_R(t) = \frac{v}{\omega_R}\sin(\omega_R t), \qquad y_R(t) = R[1 - \cos(\omega_R t)].$$

They converge on compact time intervals to the parabola, but $x_R(\tau) = v\tau - Fv\tau^3/(6mR) + O(R^{-2})$. Angular momentum about C_R is mRv, so the arc and inertial tangent both have sector area $Rv\tau/2$, with difference zero at every R. The endpoint term

$$\frac{R}{2}[x_R(\tau) - v\tau] \longrightarrow -\frac{Fv\tau^3}{12m}$$

cancels the apparent finite part. Replacing the path by its limiting parabola before subtracting the divergent reference changes this comparison. For the chord joining the **actual** arc endpoints instead,

$$A_R[q_R] - A_R[\mathrm{ch}_R] = \frac{Rv}{2}\left[\tau - \frac{\sin(\omega_R\tau)}{\omega_R}\right] \longrightarrow \frac{Fv\tau^3}{12m}.$$

This preserves precisely the matched-endpoint lens and its normalization, not the inertial tangent sector. Source/proof context: The named source passages above support the geometric comparison. Its written calculation is internal; no independent historical or proof review is claimed.

Controlled reunion of the inertial and falling arms

Keep a reference arm at zero transverse force and apply to the signal arm

$$f(t) = \begin{cases} F, & 0 < t < \tau, \\ -F, & \tau < t < 3\tau, \\ F, & 3\tau < t < 4\tau. \end{cases}$$

Both begin at $y = \dot{y} = 0$. This is branch-controlled motion; gravity needs compensation on the reference and return stages. Define $P(t) = \int_0^t f$, $Y(t) = m^{-1}\int_0^t P$. Then $P(4\tau) = Y(4\tau) = 0$; P/F is respectively t, $2\tau - t$ and $t - 4\tau$. Direct integration gives

$$\int_0^{4\tau} P^2 dt = \tfrac{4}{3}F^2\tau^3, \qquad \int_0^{4\tau} Y dt = 2F\tau^3/m, \qquad A_{\mathrm{loop}} = 2vF\tau^3/m = 12A_{\mathrm{inertial,fall}}.$$

$Y \geq 0$ peaks at $F\tau^2/m$ at 2τ. Using $m\ddot{Y} = f$ and the vanishing boundary term $mY\dot{Y}$,

$$\Delta S_{\text{loop}} = -\frac{1}{2m}\int P^2 dt = -\frac{2F^2\tau^3}{3m} = -\frac{F}{3v}A_{\text{loop}} = -\frac{4}{3}\tau\Delta E.$$

The sign and coefficient differ from the off-shell chord defect because return forces are part of this experiment. Its exact comparison with the original falling lens is $|\Delta S_{\text{loop}}| = 4FA_{\text{inertial,fall}}/v$.

Supply canonical quantum kinematics, a coherent two-arm label and $H_I = p^2/(2m)$, $H_F(t) = H_I - f(t)y$, with no additional arm-dependent scalar potentials or uncontrolled phases. In the interaction picture, $B(t) = if(t)(y + tp/m)/\hbar$ has scalar commutator

$$[B(t), B(s)] = \frac{if(t)f(s)(t-s)}{m\hbar}.$$

Higher nested commutators vanish and $\int f = \int tf = 0$ kills the first integrated term. The second yields exactly

$$U_I(4\tau)^\dagger U_F(4\tau) = \exp\left[-\frac{i}{2m\hbar}\int_0^{4\tau} P^2 dt\right] I = e^{i\Delta S_{\text{loop}}/\hbar} I.$$

The sign follows by integrating $\int f(t)\int_0^t f(s)(t-s)ds\,dt = -\int P^2 dt$. The algebra on Schwartz states extends by the forced free-particle unitaries to the motional Hilbert space. Position and momentum reunite exactly for every common initial motional preparation, not just classical centroids.

A conditional finite-copy phase threshold is not an action lattice

A balanced coherent arm label has relative phase $\phi = \Delta S_{\text{loop}}/\hbar$. Against the zero-phase reference, n independent copies and arbitrary joint binary measurement give

$$p_{\text{opt}} = \tfrac{1}{2}\left(1 + \sqrt{1 - \cos^{2n}(\phi/2)}\right).$$

On the first lobe $|\phi| \leq \pi$, success at least $p_* > 1/2$ requires

$$d_{n,p_*} = 2\hbar \arccos([4p_*(1-p_*)]^{1/(2n)}), \qquad \tau \geq \left(\frac{3md_{n,p_*}}{2F^2}\right)^{1/3}, \qquad A_{\text{inertial,fall}} \geq \frac{vd_{n,p_*}}{4F}.$$

For $n = 1, p_* = 3/4$, $d = \pi\hbar/3$. At fixed $p_* < 1$ it tends to zero as $n \to \infty$, whereas perfect discrimination requires $\pi\hbar$ for every finite n. Periodicity prevents a monotone extension beyond that lobe. The loop remains defined below the threshold; only its distinguishability at those resources decreases. The matched two-path experiment uses a different control history and retains its own noncommuting-limit tests.

Similarly, with $h_P = 2\pi\hbar$, $J\Delta\phi \geq h_P$ for N equal angular cells gives $J \geq N\hbar$, not exact quantization: $J = (N + 1/2)\hbar$ satisfies all cell bounds. Exact tiling is additional. In the conditional Kepler model $E_N = -A/N^2$, if $\omega_N = \hbar^{-1}dE_N/dN$ and $T_N = 2\pi/\omega_N$, then

$$\frac{(E_{N+1} - E_N)T_N}{h_P} = \frac{N(2N+1)}{2(N+1)^2} \longrightarrow 1.$$

The derivative rule is exact while finite differences approach it. No step transfers a comparison-path threshold to orbital cells without a physical map.

Uniform mean anomaly and a single whole-orbit action are Kepler specializations. Generic bound central motion has radial and angular actions, need not close and can have incommensurate frequencies. Reduced actions $(2\pi)^{-1}\oint p\,dq$ must be distinguished from the unreduced integrals. EBK offsets and quantum angular-momentum spectra require their own quantum

or semiclassical premises; none follows from a cell inequality. Large $J/\hbar$ describes a changing large-action family or a specified formal limit. Holding J, physical $\hbar$ and $J = N\hbar$ fixed while taking $N \to \infty$ is inconsistent.

Consequence for STATE

The inertial–parabola area, matched chord defect, receding-centre sectors and controlled reunion retain different endpoints and coefficients. Mesh size, record resolution and $\epsilon = \hbar/S_0$ are separate limits. A consistent classical geometric refinement and a supplied quantum phase benchmark are both constructed here; independent exclusion of zero action remains open.